pyALDIC Quick Guide

Two-page reference. Full walkthrough: `user-guide.tex`.

1. Install & launch

Pick one install path (all are equivalent at runtime):

- **From PyPI** (easiest): `pip install al-dic`.
- **From a GitHub Release wheel**: download `al_dic-*.whl` from github.com/zachtong/pyALDIC/releases and run `pip install al_dic-*.whl`. Useful when behind a firewall that blocks PyPI.
- **From source** (development): `git clone https://github.com/zachtong/pyALDIC.git` then `pip install -e ".[dev]"` inside the clone.

Requires Python $\geq$ 3.10. Launch: `al-dic` (or `python -m al_dic`).

2. Load images

Drop an image folder onto the drop zone or click Browse. All files must be the same size. Frame 1 is always the reference. Frames stream from disk on demand, so hundreds of 4K images load without exhausting RAM.

3. Choose Workflow Type

This decides which frames need a Region of Interest, so do it *before* drawing.

Experiment	Tracking + Ref Update
Soft material <2% strain	Accumulative
2–30% strain, any rotation >5°, impact	Incremental + Every Frame
Known quasi-static load steps	Incremental + Custom Frames

Solver: **AL-DIC** default (smoother, $\sim 3\times$ slower); **Local DIC** for quick iteration.

4. Draw Region of Interest

The blue-accent hint above the toolbar tells you which frames need a region. Toolbar:

- **+ Add**, **Cut**, **+ Refine** (brush, frame 1 only). Each opens a Polygon / Rectangle / Circle menu.
- **Import** / **Batch Import** – load PNG masks.
- **Save** / **Invert** / **Clear** – per- frame mask operations.

For per-frame editing, click the Need / Edit / Add button on a row in the image list.

5. Parameters

Control	Guidance
Subset Size	41 default; larger (51–81) for noise, smaller (21–31) for fine speckle.
Subset Step	8 for 512^2, 16 for 1024^2. Halving step $\rightarrow \sim 4\times$ runtime.
Search Range	20 default; raise for large inter-frame motion. Auto-expand is on.
Refinement	Check Inner / Outer to densify mesh near hole or ROI boundaries; pick a level (Light to Ultra).

Advanced section (collapsed by default) holds the initial-guess mode and auto-expand toggle; defaults are fine.

Starting Points (special init-guess mode). Select this radio for large displacement (> 50 px) or discontinuous fields (cracks, shear bands). Every connected mask region shows as yellow on the canvas; place at least one point per region (*Place Starting Points* to click, or *Auto-place* to pick the highest- NCC node per region) until every region turns green. Run stays disabled until that's true. On a reference-frame switch the points are warped automatically; if warping fails the pipeline auto-places fresh points on the new reference and continues.

6. Run & Cancel

Click the blue **Run DIC Analysis**. Progress, elapsed / remaining, and console log appear on the right. Red **Cancel** stops cleanly (partial results kept).

Fatal errors raise a modal dialog; the full traceback is in the console log.

7. View results

On the right sidebar after a run:

- **FIELD**: Disp U / Disp V toggle. *Show on deformed frame* checkbox sits here (controls WHERE, not how).
- **VISUALIZATION**: Colormap, Auto / Fixed range radio pair (Fixed = manual Min / Max), opacity. Numeric boxes accept both "." and "," decimals.
- **PHYSICAL UNITS**: Pixel size + frame rate – scales displacement / velocity output; strain is dimensionless.

8. Strain post-processing

A separate **Strain window** opens automatically when the run finishes.

- **Method**: *Plane fitting* (smooth, default) vs *FEM nodal* (higher resolution, noisier).
- **VSG size**: Virtual strain gauge diameter (default 41 px).
- **Smoothing**: Off / Light ($0.5\times$ step) / Medium ($1\times$ step) / Strong ($2\times$ step) ▲.
- **Strain type**: **Infinitesimal** (small deformation only), *Eulerian-Almansi*, **Green-Lagrangian** (**use for any rotation > 2°**; it is exactly zero under rigid rotation).

Click green **Compute Strain** whenever parameters change. Field selector now exposes `strain_exx`, `strain_eyy`, `strain_exy`, principal strains, `strain_maxshear`, `strain_von_mises`, `strain_rotation`, plus the displacement fields.

9. Export

Both windows have an **Export Results** button; they open the same 5-tab dialog:

- **Data**: NPZ / MAT / CSV of any selected fields.
- **Images**: per-field JPEG (quality 60–100) / PNG / TIFF per frame; resolution presets cap the long edge (512–2048 px or full). Colorbar ON by default.
- **Animation**: MP4 / GIF, encoded frame-by-frame (no

RAM spike); *Frame step* keeps every Nth frame with playback duration preserved. Requires `ffmpeg` on PATH for MP4 (falls back to GIF otherwise).

- **Report**: multi-page PDF summary of the run.
- **Preview & Colorbar**: WYSIWYG preview through the real export path. Colorbar position / font / thickness / background / margin; field colormap–range–opacity synced two-way with Images; *Apply to all fields*.

10. Save / resume sessions

File → **Save Session** (`Ctrl+S`) writes a single `.aldic` bundle: every Region of Interest, all parameters, view state, and — optionally, a prompt shows the size estimate — the computed results. Async, with a progress dialog.

File → **Open Session** (`Ctrl+O`) restores the exact page (frame, field, colormap, ranges) with no recompute. Legacy `.aldic.json` files still open. If the image folder is gone, parameters and Regions of Interest still load; re-drop the images manually. Starting Points are not saved — re-place them before the next Run.

Windows: **File** → **Associate .aldic files** makes double-clicking a session file open it in pyALDIC.

11. Troubleshooting quick table

Symptom	Likely cause / fix
Run button disabled	No images or no Region of Interest on frame 1.
Crazy strain under big rotation	Using Infinitesimal strain; switch to Green-Lagrangian.
Strain map / export all blank or NaN	Plane-fit edge-trim removed every node — ROI too thin or VSG too large. Lower *Trim* (α), reduce VSG size, or use FEM nodal.
FFT peaks clipped warnings	Raise Search Range or keep auto-expand on.
Refine brush grayed out	Brush is frame-1 only; navigate to frame 1.
Results poor near hole edges	Enable *Refine Inner Boundary*.
Slow run	Halve Subset Step for next iteration; use Local DIC for previews.
Session won't load	Check console: corrupt file, future schema, or missing image folder (warning only – params restore).